

Cognitive-AI Approaches to Adaptive Scheduling in Braille Memory Retention Using Hybrid SM-2 Models

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Abstract—The spaced repetition algorithms are effective in the long term memory retention but current approaches tend to sacrifice interpretability to personalization. In this study, the hybrid scheduling algorithm, which unites the transparent and rule-based SM-2 with machine learning-based adjustments to personal learning behavior, is proposed. The hybrid approach with a IoT-enabled tactile flashcard applied to Braille literacy education, the hybrid methodology attains a similar level of learning performance as the pure machine learning approaches with a large reduction of the long-term review effort. Findings of a one-year assessment suggest that the hybrid scheduler is highly recall efficient with significantly reduced daily review burden as compared to a fully data-driven model. These results indicate the strengths of integrating algorithmic transparency with adaptive learning to educational systems that need sustainable long-term retention.

Index Terms—spaced repetition, adaptive learning, hybrid scheduling, SM-2 algorithm, machine learning, Braille literacy, memory retention, educational technology

I. INTRODUCTION

Braille literacy is a necessary but underdeveloped skill of people with the visual impairment but only 10-25 percent of legally blind people in the United States are able to ensure proficiency in it [1]. Since mastering the Braille requires discrimination of tactile patterns and continued practice, the spaced repetition procedures which reduce memory deterioration are especially effective in this respect [2], [3]. However, more traditional methods like SM-2 [3], use inflexible scheduling heuristics that ignore the inter-individual variability but machine learning schedulers like FSRS [5] create interpretative ambiguity and over-parameterization vulnerability to over-fitting. The exploration is a hybrid spaced repetition framework that combines data-driven adaptivity and rule-directed transparency. We describe a Hybrid SM-2+AI scheduler with adjustable interpolation of algorithmic and learned components with an appraisal apparatus and considerations of implementation of assistive Braille pedagogy.

II. BACKGROUND AND RELATED WORK

A. Spaced Repetition Theory

The spacing effect demonstrates that information reviewed at increasing intervals is retained more effectively than massed

practice [2]. This phenomenon has been replicated across diverse domains establishing spaced repetition as one of the most robust findings in cognitive psychology [10], [12], [17]. Ebbinghaus’s forgetting curve (Figure 1) reveals exponential

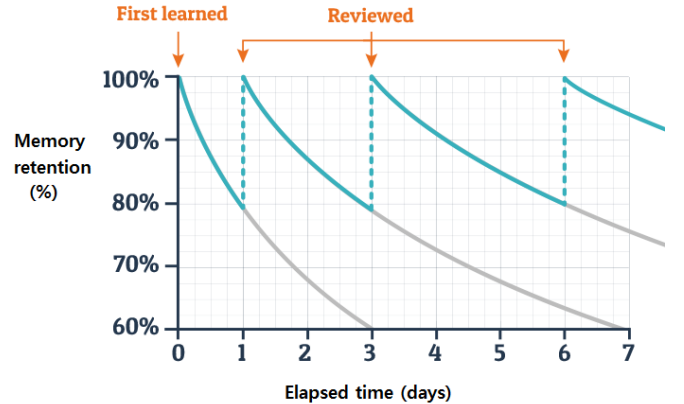


Fig. 1. Memory retention showing forgetting curve (gray) and spaced repetition effect (teal). Strategic reviews (arrows) maintain high retention and flatten forgetting. Adapted from Ebbinghaus [2]

memory decay with retention dropping to 60 percent within one day and 70-75 percent over subsequent days [9]. Strategic review interventions reset this decay maintaining retention near 100 percent. This forms the computational foundation for scheduling algorithms where reviews must occur before memory strength falls below a critical threshold with intervals expanding as memory consolidates. Scheduling algorithms determine optimal review intervals balancing retention maximization, review burden minimization and adaptation to individual differences. Meta-analyses across 271 experiments show spaced practice improves learning outcomes by 40-70 percent compared to massed practice with benefits in 96 percent of studies [11], [16].

B. The SM-2 Algorithm

The SuperMemo-2 (SM-2) algorithm represents an early computational model for spaced repetition [3]. SM-2 maintains two state variables per flashcard: the *ease factor* (EF),

representing item difficulty where higher values indicate easier items requiring less frequent review, and the *repetition count*, tracking successful recalls.

The algorithm operates on a quality score q (0-5) where 0 indicates complete recall failure and 5 represents perfect recall.

1) *Ease Factor Update Mechanism:*

$$EF' = \max(1.3, EF + 0.1 - (5 - q)(0.08 + 0.02(5 - q))) \quad (1)$$

The term $(5 - q)$ quantifies deviation from perfect recall. The constant 0.08 applies base penalty for suboptimal performance, $0.02(5 - q)$ introduces quadratic penalty escalating with declining scores, and 0.1 provides positive adjustment offsetting penalties.

Three behavioral regimes emerge: perfect recall ($q = 5$) yields $EF' = EF + 0.1$ gradually increasing ease; near-perfect ($q = 4$) produces minimal change; suboptimal ($q \leq 3$) substantially decreases ease and shortens intervals. The constraint $EF' \geq 1.3$ prevents excessive interval compression.

2) *Interval Scheduling:* The first two intervals are fixed at $I_1 = 1$ day and $I_2 = 6$ days, while subsequent intervals scale multiplicatively:

$$I_n = I_{n-1} \times EF', \quad \text{for } n > 2. \quad (2)$$

This exponential spacing reflects that successfully recalled items require progressively longer intervals. When performance falls below threshold ($q < 3$), the algorithm resets repetition count to zero and reverts to one day, restarting the learning sequence.

3) *Limitations and Contemporary Perspectives:* SM-2 exhibits several limitations: every item is initialized with the same ease factors, the constant hand-crafted update rule offers limited personalization and the single dimensional ease factor fails to represent multidimensional memory processes. Modern studies have shown that machine learning methods are capable of better predicting the learning patterns and schedules of individual learners and optimizing them with improved precision. [4].

C. Machine Learning Approaches to Spaced Repetition

More recent developments use machine learning to forecast memory dynamics and solve scheduling more intelligently than rule-based algorithms by using massively scaled student data to learn personalized memory models.

1) *Half-Life Regression:* Settles and Meeder [7] introduced half-life regression (HLR), combining psycholinguistic theory with machine learning. The model builds on the Ebbinghaus forgetting curve where memory decays exponentially:

$$p = 2^{-\Delta/h}, \quad (3)$$

where p denotes recall probability, Δ represents lag time since last practice, and h is the half-life measuring memory trace strength. When $\Delta = h$, recall probability drops to $p = 0.5$.

HLR estimates half-life as an exponential function of learner features:

$$\hat{h}_\Theta = 2^{\Theta \cdot x}, \quad (4)$$

where x represents a feature vector summarizing student's previous exposure and Θ contains learned weights. Features include interaction counters (exposures, correct recalls, errors) and lexeme-specific indicators capturing inherent difficulty.

The training objective minimizes L2-regularized squared loss:

$$\Theta^* = \arg \min_{\Theta} \sum_{i=1}^D \left[(p - \hat{p}_\Theta)^2 + \alpha(h - \hat{h}_\Theta)^2 + \lambda \|\Theta\|_2^2 \right], \quad (5)$$

where α controls half-life prediction importance and λ provides regularization.

HLR reduced prediction error by 45% compared to Leitner and Pimsleur baselines on 12.9 million Duolingo learning traces. Interaction features alone achieved comparable accuracy while improving user engagement by 12% in production.

2) *Stochastic Optimization and MDP Formulations:* Ye et al. [8] frame scheduling as stochastic shortest path optimization, formulating reviews as actions within a Markov decision process balancing retention maximization and review cost minimization through reinforcement learning.

3) *The FSRS Algorithm and DSR Memory Model:* The Free Spaced Repetition Scheduler (FSRS) implements the DSR model tracking three memory components [5]:

- **Difficulty (D):** Intrinsic material complexity where higher values correspond to smaller stability increases upon review.
- **Stability (S):** Memory strength measured in days. Higher stability means slower forgetting. Increases after successful recall, decreases after failures.
- **Retrievability (R):** Probability of successful recall at a given moment depending on elapsed time and current stability. Decays continuously over time according to the forgetting curve.

The system models retrievability as exponential decay:

$$R(t) = \exp\left(-\frac{t}{S \cdot 9}\right), \quad (6)$$

where t represents elapsed time since last review and S is stability. When $t = S \cdot 9$, retrievability drops to approximately 90%.

FSRS-5 utilizes 17 parameters (FSRS-6 uses 19 parameters) governing DSR dynamics, trained using gradient descent optimization on review data. For users with insufficient history, default parameters derived from analyzing millions of reviews are employed. FSRS reduces reviews by 20-30% compared to SM-2 while maintaining retention levels [5].

4) *Challenges and Trade-offs in ML-Based Scheduling:* ML methods have better adaptation, but have practical issues such as large computational costs, risk of overfitting with small data, and uninterpretability. The 21 parameter model of FSRS that works based on three interacting variables does not provide much intuitive information on decision-making in terms of scheduling. This lack of opaqueness, as opposed to the transparent updates on SM-2, destroys user trust and the development of metacognitive awareness, making this a black

box system instead of allowing the learners to learn how to successfully review.

D. Braille Learning as Pattern Recognition

Braille literacy involves learners having to be able to quickly recognize the dots raised in a 2x3 grid and be able to encode letters, numbers or symbols using tactile sensing.

Braille reading is a perceptual learning (the ability to distinguish between patterns) and memory consolidation (associating patterns with the meaning). Spaced repetition confronts directly the problem of consolidation, which could speed up fluency and lessen the cognitive load in the early stage of learning [17].

III. PROBLEM STATEMENT

This research addresses: *Can a hybrid scheduling algorithm combining rule-based SM-2 logic with machine learning adaptation achieve comparable learning outcomes to pure ML approaches while maintaining lower computational costs and greater interpretability?*

We identify three limitations in existing approaches. SM-2 is transparent and computationally simple but static, unable to adapt to individual learning patterns beyond generic ease factor adjustments. ML-based approaches are adaptive and personalizable but opaque, prone to overfitting, and lacking transparency for learners or educators to understand scheduling rationale. Neither approach balances long-term retention, daily review burden, and interpretability required for real-world applications.

Our hypothesis is that a hybrid approach achieves the best of both paradigms: leveraging SM-2's interpretable foundation while incorporating ML-based personalization to improve scheduling without sacrificing transparency or imposing excessive computational demands.

IV. METHODOLOGY

A. Dataset Details and Preprocessing

This research utilizes two established datasets commonly employed in spaced repetition studies. The first dataset comprises review histories from 20,000 Anki users containing millions of review entries with timestamps, quality ratings, and card identifiers (FSRS-Anki-20k, HuggingFace). The second dataset provides additional learner logs from the Harvard Dataverse Spaced Repetition collection, offering cross-validation data to assess generalization.

1) *Raw Dataset Format:* Table I shows a sample of the raw FSRS-Anki dataset format. Each row represents a single review event with card identifier, review number, elapsed time measurements, user rating (1-4), card state (0=new, 1=learning, 2=review), and review duration in milliseconds.

2) *Dataset Preprocessing:* Data preprocessing involves filtering incomplete review sequences, normalizing timestamps to daily intervals, and removing outliers with intervals exceeding 365 days. Raw review events are aggregated into review sessions and transformed to compute derived features

TABLE I
SAMPLE RAW FSRS-ANKI DATASET

card_id	review_th	elapsed_days	rating	state	duration
2822	88112	92	3	2	8086
2823	40841	-1	1	0	60000
2823	40845	0	3	1	17028
2823	40857	0	3	1	24873
2823	40869	1	3	2	10271

including half-life, recall probability, difficulty, and review history sequences.

Table II shows the normalized format after initial preprocessing. Each row represents a review session with computed half-life (h), repetition index (i), predicted recall probability (p_{recall}), time since last review (δt), review history string ($r_{history}$), interval history ($t_{history}$), probability history ($p_{history}$), difficulty (d), and total review count.

TABLE II
NORMALIZED DATASET FORMAT

h	i	p_{recall}	Δt	$r_{history}$	$t_{history}$	$p_{history}$	d
1.5	2	0.63	1	1	0	0.50	4
1.5	3	0.16	4	1,1	0,1	0.50,0.63	4
6.0	4	0.50	6	1,1,1	0,1,4	0.50,0.63,0.16	4

Final preprocessing augments each record with temporal context features for machine learning models. Table III shows the final format including group count for cross-validation, last half-life (h_{prev}), and last recall probability (p_{prev}) for sequential prediction tasks.

TABLE III
FINAL PREPROCESSED DATASET FORMAT

h	i	p_{recall}	Δt	d	h_{prev}	p_{prev}
1.5	2	0.16	4	1	-	-
1.5	2	0.63	1	10	-	-
1.5	2	0.16	4	10	-	-
1.5	2	0.63	1	10	-	-

Review histories are segmented into training (80%) and testing (20%) splits while maintaining temporal ordering to prevent data leakage. Both datasets are mapped to the SSP-MMC-Plus schema for consistent processing across all scheduling algorithms.

3) *Dataset Limitations and Generalizability:* Even though both datasets are comprised of millions of review logs, they are not related to Braille learners but to the general learners of Anki. This raises the possibility of generalizability issues because learning Braille has special cognitive processes of recognizing tactile patterns that would not be the same as learning using visual flashcards.

Although this is the case, there is a lot of cognitive overlap between modalities. The two need the memorization of separate symbolic units, the association of symbols and language meanings, the distinction between similar patterns, and the spacing effect in order to enhance memory traces. Braille is

a haptic code system of cells to letters/phonemes (alphabet) like letters of an alphabet. Nevertheless, the consolidation of memory underlying this is similar forgetting curves across sensory modalities, and it has already been shown that spacing effects are generalized across content and across modalities. [17].

These dataset limitations imply our findings represent a conservative estimate of hybrid scheduling effectiveness for Braille literacy. Further improvements may emerge when optimized using tactile Braille learner data.

B. Models and Techniques

We implement and evaluate three scheduling policies to compare traditional, machine learning, and hybrid approaches.

1) *Baseline SM-2*: Standard implementation following Wozniak’s original specification with ease factor updates based on quality scores and exponentially growing intervals.

2) *Pure-ML Scheduler*: An ensemble of three machine learning models for half-life prediction:

Half-Life Regression (HLR) uses logistic regression with features $\sqrt{+n_{right}}$, $\sqrt{1+n_{wrong}}$, and bias term, where n_{right} and n_{wrong} count successful and failed reviews [7]. The model predicts:

$$h_{HLR} = 2^{\theta_0 + \theta_1 \sqrt{1+n_{right}} + \theta_2 \sqrt{1+n_{wrong}}} \quad (7)$$

where θ_i are learned weights.

Difficulty-Half-life Prediction (DHP) is a sequential model tracking item difficulty d (initialized from error count, capped at 18) and updating half-life based on recall success:

$$h' = \begin{cases} \exp(r_a) \cdot d^{r_b} \cdot h^{r_c} \cdot (1-p)^{r_d} & \text{if recall} \\ \exp(f_a) \cdot d^{f_b} \cdot h^{f_c} \cdot (1-p)^{f_d} & \text{if forget} \end{cases} \quad (8)$$

where $p = 2^{-\Delta t/h}$ is predicted recall probability, h is current half-life, and $r_a, r_b, r_c, r_d, f_a, f_b, f_c, f_d$ are learned parameters [6].

RNN-HLR processes review history sequences (r_i, t_i, p_i) to predict half-life, capturing temporal dependencies through GRU architecture followed by a fully connected output layer [6].

The ensemble combines predictions through confidence-weighted averaging. The predicted half-life h_{ML} determines the review interval as $I_{ML} = \lfloor h_{ML}/1.5 \rfloor$ to target approximately 90% recall probability.

3) *Hybrid SM-2+AI Scheduler*: Our proposed approach combines SM-2 and ML predictions through an adaptive blending mechanism:

The adaptive blending factor β adjusts based on review history:

$$\beta(n) = \begin{cases} 0.0 & \text{if } n < 3 \\ \min(0.0 + (n-3) \cdot 0.1, 0.8) & \text{otherwise} \end{cases} \quad (9)$$

where n is the review count. This interpolates from pure SM-2 for new items ($\beta = 0$) toward ML-influenced scheduling for established items ($\beta \leq 0.8$).

Algorithm 1: Hybrid Scheduling Algorithm

Input: user_id, item_id, rating, timestamp

Output: interval I , recall probability p_{recall}

$I_{SM2}, h_{SM2} \leftarrow \text{SM2.schedule}(\text{rating});$

$state \leftarrow \text{get_state}(\text{user_id}, \text{item_id});$

$\beta \leftarrow \text{calculate_adaptive_beta}(state);$

if $\beta > 0$ **and** $state.repetitions \geq 3$ **then**

$(I_{ML}, h_{ML}, conf) \leftarrow \text{predict_with_ml}(state);$

if $I_{ML} \neq \text{None}$ **and** $conf > 0.1$ **then**

$I_{hybrid} \leftarrow \lfloor (1-\beta) \cdot I_{SM2} + \beta \cdot I_{ML} \rfloor;$

$h_{hybrid} \leftarrow (1-\beta) \cdot h_{SM2} + \beta \cdot h_{ML};$

else

$I_{hybrid} \leftarrow I_{SM2}, h_{hybrid} \leftarrow h_{SM2};$

$I_{hybrid} \leftarrow I_{SM2}, h_{hybrid} \leftarrow h_{SM2};$

$p_{recall} \leftarrow 2^{-I_{hybrid}/h_{hybrid}};$

return $I_{hybrid}, p_{recall};$

The blending factor is modulated by review consistency:

$$\beta_{adjusted} = \beta_{base} \cdot \left(0.7 + 0.3 \cdot \frac{\bar{r} - 2}{3} \right) \quad (10)$$

where $\bar{r}$ is the average rating over the last 5 reviews.

C. Evaluation Techniques

1) *Memory Half-Life Concept*: We adopt memory half-life as the central evaluation concept, defined as the time until recall probability falls to 0.5. We model recall probability using exponential forgetting:

$$P(t) = 2^{-t/h} \quad (11)$$

where t is time since last review and h is the half-life parameter. For SM-2, we estimate half-life as $h_{SM2} \approx I_{SM2}/1.5$ assuming 90% target recall. For ML models, half-life is directly predicted from review history features.

2) *Evaluation Metrics*: We evaluate schedulers using the following metrics over a 365-day simulation:

Recall Curves measure average recall probability at checkpoints (30, 60, 180, 365 days). THR (Target Half-life Reached) calculates the percentage of items achieving half-life ≥ 180 or 365 days. SRP (Summation of Recall Probability) measures total system retention capacity as $SRP = \sum_{i=1}^N p_i$ where p_i is predicted recall probability for item i . WTL (Words Total Learned) counts items with recall $\geq 90\%$ for last three consecutive reviews. Daily Cost measures average reviews per day. Efficiency is calculated as $\eta = SRP/\text{Daily Cost}$.

D. Research Flow Overview

Figure 2 illustrates the research pipeline. We merge FSRS-Anki-20k and Harvard Dataverse datasets, extract features, train SM-2 baseline and ML models (HLR, DHP, RNN-HLR), then integrate them through adaptive blending in the Hybrid scheduler.

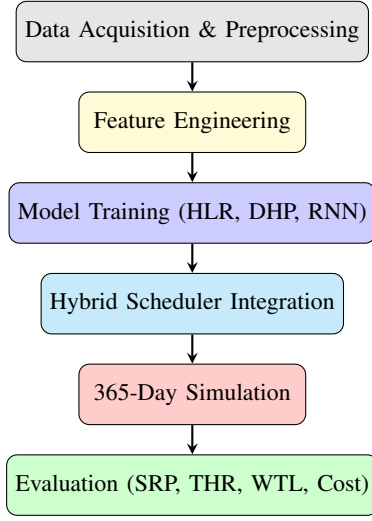


Fig. 2. Research methodology pipeline: data preprocessing, model training, hybrid integration, simulation, and evaluation.

1) *Simulation Framework*: We implement a discrete-event simulator managing 20,000 flashcards over 365 days with daily budget of 600 cost units. Review costs are 3 units for successful recall, 9 units for failed recall, and 12 units for new items. Each card tracks difficulty (1-10), half-life, recall probability, repetition count, lapse count, and review history.

Daily processing handles due reviews first, then introduces new items until reaching budget limit. Recall success is determined stochastically using $p = 2^{-t/h}$ where t is time since last review. Cards achieving half-life ≥ 360 days are considered mastered and removed from rotation. The simulator tracks SRP, THR, WTL, and daily cost for direct comparison under realistic constraints.

V. RESULTS

Table IV presents comprehensive results comparing the three scheduling approaches across all evaluation metrics over 365 days.

TABLE IV
KEY METRICS AT DAY 365

Metric	SM-2	Hybrid	Pure-ML
WTL	300	305	178
Daily Cost	20.0	21.7	166.5
Efficiency	0.365	0.375	0.389
SRP	2661	2970	3964
Total Reviews	7284	7926	10155+

A. Words Total Learned (WTL)

WTL measures items achieving stable mastery ($\geq 90\%$ recall for three consecutive reviews). Pure-ML achieves rapid initial learning reaching 274 items by day 7 but plateaus at 178 items by day 30. SM-2 and Hybrid show delayed onset (0 items through day 141) but accelerate dramatically afterward, reaching 300 and 305 items respectively by day 365 (Figure 3).

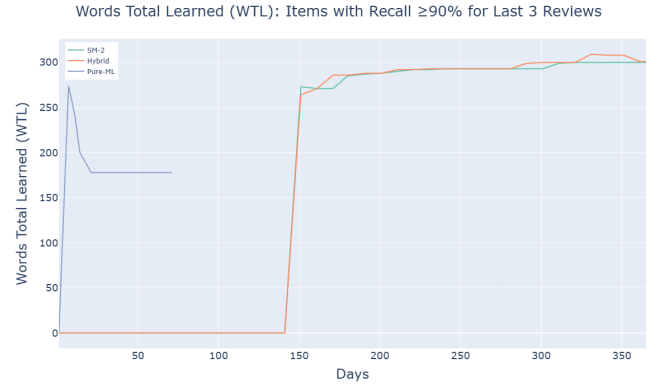


Fig. 3. Words Total Learned over 365 days. Hybrid and SM-2 achieve 300+ mastered items compared to Pure-ML (178).

This suggests Pure-ML optimizes for short-term recall at the expense of long-term consolidation while Hybrid prioritizes sustainable mastery without compromising effectiveness.

B. Daily Review Cost

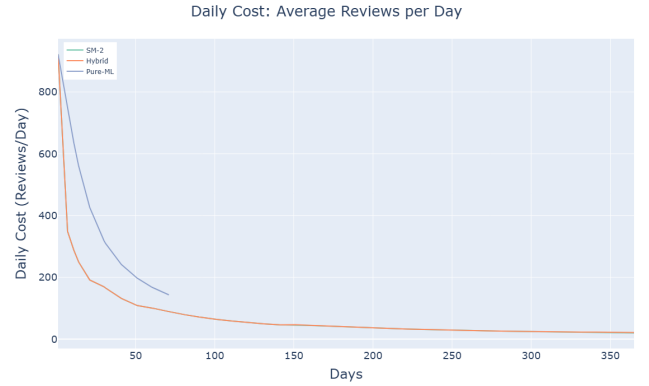


Fig. 4. Daily Cost over time. Pure-ML maintains high burden (166.5/day) while Hybrid reduces to 21.7/day (87% reduction).

Daily review burden shows dramatic differences (Figure 4). Pure-ML maintains 166.5 reviews/day by day 365, prohibitively high for practical deployment. Hybrid reduces to 21.7 reviews/day (87% reduction versus Pure-ML), nearly matching SM-2's 20.0 reviews/day. Pure-ML's aggressive scheduling results in 10,155 total reviews compared to Hybrid's 7,926 reviews (22% reduction). This demonstrates a key advantage: preventing review explosion that undermines Pure-ML's practical viability.

C. Efficiency and Recall

Efficiency metrics reveal how effectively each scheduler converts review effort into retention (Figure 5). Pure-ML achieves highest initial efficiency (0.54 at day 1) but declines to 0.39 by day 365. Hybrid stabilizes at 0.37 by day 365, nearly matching Pure-ML despite vastly lower review costs. This indicates Pure-ML's additional reviews do not translate to proportionally greater retention.



Fig. 5. Efficiency (SRP ÷ Cost) showing Hybrid achieves comparable final efficiency (0.37) to Pure-ML (0.39) despite lower review burden.

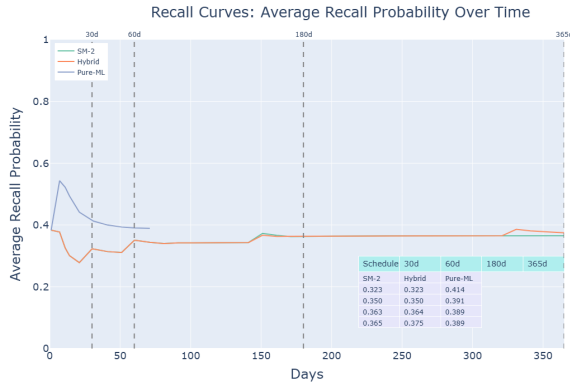


Fig. 6. Recall Curves at key checkpoints. All approaches converge to similar recall levels long-term.

Average recall probability (Figure 6) shows convergence across all approaches at longer time scales, with Pure-ML at 0.389, Hybrid at 0.375, and SM-2 at 0.365 by day 365, suggesting different strategies ultimately achieve similar steady-state recall probabilities.

D. Summation of Recall Probability (SRP)

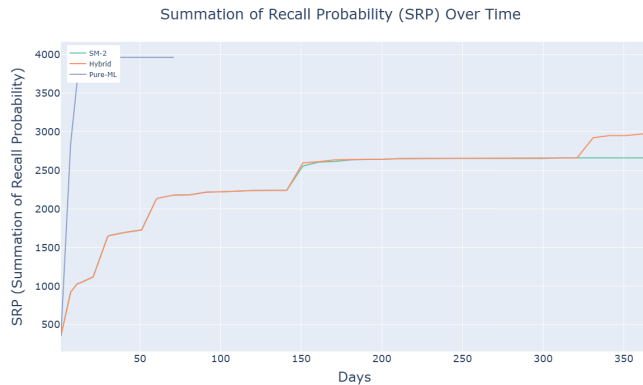


Fig. 7. SRP showing Pure-ML plateaus early while Hybrid shows sustained growth to day 365.

SRP measures total system retention capacity (Figure 7). Pure-ML achieves highest SRP early (3,963 by day 30) but shows minimal growth thereafter. Hybrid demonstrates continuous growth from 1,642 at day 30 to 2,970 at day 365, indicating sustained learning progress rather than early plateau.

E. Target Half-Life Reached (THR)



Fig. 8. THR at 180 and 365 days. Pure-ML achieves 13.8% at 180-day half-life but plateaus; SM-2 and Hybrid show 0%.

Neither SM-2 nor Hybrid achieves items with half-life ≥ 180 days while Pure-ML reaches 13.8% at 180-day half-life by day 7 but plateaus (Figure 8). All approaches have room for improvement in extending memory durability to very long time scales, though Pure-ML's early advantage does not translate to superior long-term mastery.

VI. DISCUSSION

A. Interpretation of Results

Our results demonstrate that the Hybrid SM-2+AI scheduler successfully balances long-term retention, review burden, and interpretability through three key findings:

First, hybrid approaches achieve comparable or superior long-term outcomes. The Hybrid scheduler's 305 mastered items at day 365 matches SM-2 (300) and far exceeds Pure-ML (178), despite Pure-ML's initially superior recall probabilities. This suggests Pure-ML's aggressive early scheduling (745 reviews/day at day 7) may interfere with memory consolidation processes that benefit from temporal spacing. The adaptive β blending factor allows Hybrid to inherit SM-2's conservative early scheduling while gradually incorporating ML personalization as review history accumulates.

Second, hybrid approaches dramatically reduce review burden. The 87% reduction in daily reviews (21.7 versus 166.5 at day 365) makes Hybrid practically deployable in real-world contexts where learner time is limited. This is achieved through conservative $\beta_{max} = 0.8$ limiting ML influence, gradual β growth requiring 3 reviews before ML incorporation, and consistency-based modulation reducing β when recent ratings show high variance. Pure-ML's demands would likely lead to burnout or abandonment in actual use.

Third, the blending mechanism provides interpretability. The linear formula $I_{hybrid} = (1 - \beta) \cdot I_{SM2} + \beta \cdot I_{ML}$ preserves transparency, allowing educators and learners to understand scheduling rationale. Ensemble averaging of HLR, DHP, and RNN predictions provides robustness with graceful degradation to pure SM-2 when ML models fail.

B. Implications for Braille Education

These results imply the use of Hybrid scheduling as IoT-enabled tactile flashcards as an effective, sustainable learning pathway to Braille literacy. The light load of reviewing enables the daily practice and retains the spaced repetition advantages that hastens the mastering of patterns. Clear scheduling principles enable the instructors to clarify when to review, which lead to metacognitive understanding of the memory processes.

This is supported by the fact that the Hybrid approach performs well in similarity to SM-2 which confirms that the augmentation of ML does not hurt results in cases of imperfect predictions. This strength is essential in the context of assistive technologies in which the reliability of the algorithm has a direct influence on the success of the learners.

C. Limitations and Future Directions

This study has several limitations. The datasets originate from visual flashcard learners rather than Braille learners, potentially limiting generalizability to tactile learning contexts. The simulation uses fixed cost parameters that may not reflect actual cognitive demands of Braille pattern recognition. Additionally, the 365-day simulation period may be insufficient to capture very long-term retention dynamics.

Future work should:

- 1) Validate findings with actual Braille learners using tactile flashcard prototypes
- 2) Replace black-box ML models with interpretable alternatives (decision trees, rule lists) that preserve transparency while maintaining adaptability
- 3) Extend to multi-modal learning combining tactile Braille with audio reinforcement
- 4) Develop efficient implementations for resource-constrained IoT devices with on-device learning capabilities

VII. CONCLUSION

This research demonstrates that hybrid scheduling algorithms combining rule-based SM-2 logic with machine learning adaptation achieve the benefits of both paradigms: interpretability and efficiency from SM-2, plus personalization from ML. Applied to Braille literacy education, the Hybrid SM-2+AI scheduler achieves long-term mastery outcomes matching or exceeding both baseline and pure ML approaches while reducing daily review burden by 87% compared to Pure-ML.

These findings validate the hybrid approach as a promising direction for educational technologies requiring sustained retention with manageable learner burden. The explicit blending mechanism provides transparency that supports trust and adoption, addressing a critical barrier to ML deployment in education.

For Braille literacy specifically, IoT-enabled tactile flashcards using Hybrid scheduling offer a practical pathway to accelerate learning and improve outcomes for individuals with visual impairments. By balancing algorithmic transparency

with adaptive personalization, such systems can provide effective support for skill acquisition that directly impacts employment and educational opportunities.

The broader lesson—that hybrid human-AI systems can outperform purely algorithmic or purely learned approaches—has implications beyond spaced repetition to any educational technology requiring interpretability, personalization, and practical deployability.

REFERENCES

- [1] The Stanford Daily, “Braille literacy rates and employment outcomes among individuals with visual impairments,” accessed 2024.
- [2] H. Ebbinghaus, *Memory: A Contribution to Experimental Psychology*. New York, NY, USA: Teachers College, Columbia University, 1913.
- [3] J. Jankowski, “Application of a computer to improve the results obtained in working with the SuperMemo method,” SuperMemo Blog, Nov. 2022.
- [4] J. Xie, “The Role of Reinforcement Learning in Enhancing Education: Applications in Psychological Education and Intelligent Tutoring Systems,” *Highlights in Science, Engineering and Technology*, vol. 124, pp. 123–131, Feb. 2025. doi:10.54097/rkxbvk42.
- [5] L. Ye *et al.*, “FSRS: Free Spaced Repetition Scheduler,” GitHub repository, 2023. [Online]. Available: <https://github.com/open-spaced-repetition/free-spaced-repetition-scheduler>
- [6] J. Su, J. Ye, L. Nie, Y. Cao, and Y. Chen, “Optimizing spaced repetition schedule by capturing the dynamics of memory,” *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 10, pp. 10085–10097, Oct. 2023.
- [7] B. Settles and B. Meeder, “A Trainable Spaced Repetition Model for Language Learning,” in *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 1848–1858, Jan. 2016. doi:10.18653/v1/P16-1174.
- [8] J. Ye, J. Su, and Y. Cao, “A stochastic shortest path algorithm for optimizing spaced repetition scheduling,” in *Proc. 28th ACM SIGKDD Conf. Knowl. Discovery Data Mining*, 2022, pp. 4381–4390.
- [9] J. M. J. Murre and J. Dros, “Replication and analysis of Ebbinghaus’ forgetting curve,” *PLOS ONE*, vol. 10, no. 7, p. e0120644, Jul. 2015, doi: 10.1371/journal.pone.0120644.
- [10] N. J. Cepeda, H. Pashler, E. Vul, J. T. Wixted, and D. Rohrer, “Distributed practice in verbal recall tasks: A review and quantitative synthesis,” *Psychological Bulletin*, vol. 132, no. 3, pp. 354–380, 2006, doi: 10.1037/0033-2909.132.3.354.
- [11] G. M. Donoghue and J. A. C. Hattie, “A meta-analysis of ten learning techniques,” *Frontiers in Education*, vol. 6, p. 581216, 2021, doi: 10.3389/educ.2021.581216.
- [12] H. P. Bahrick, L. E. Bahrick, A. S. Bahrick, and P. E. Bahrick, “Maintenance of foreign language vocabulary and the spacing effect,” *Psychological Science*, vol. 4, no. 5, pp. 316–321, Sep. 1993, doi: 10.1111/j.1467-9280.1993.tb00571.x.
- [13] M. L. Knabe and H. A. Vlach, “When are difficulties desirable for children? First steps toward a developmental and individual differences account of the spacing effect,” *Journal of Applied Research in Memory and Cognition*, vol. 9, no. 4, pp. 447–454, 2020, doi: 10.1016/j.jarmac.2020.07.007.
- [14] B. Ma, S. Fukui, Y. Ando, and S. Konomi, “Personalized language learning using spaced repetition scheduling,” in *Proc. 2025 Int. Conf. Artificial Intelligence in Education*, 2025, isbn: 978-3-031-98458-7, doi: 10.1007/978-3-031-98459-4_19.
- [15] T. P. Novikoff, J. M. Kleinberg, and S. H. Strogatz, “Education of a model student,” *Proc. Natl. Acad. Sci.*, vol. 109, no. 6, pp. 1868–1873, Feb. 2012, doi: 10.1073/pnas.1109863109.
- [16] S. K. Carpenter, N. J. Cepeda, D. Rohrer, S. H. K. Kang, and H. Pashler, “Using spacing to enhance diverse forms of learning: Review of recent research and implications for instruction,” *Educational Psychology Review*, vol. 24, no. 3, pp. 369–378, Sep. 2012, doi: 10.1007/s10648-012-9205-z.
- [17] J. Kornmeier, M. Spitzer, and Z. Sosic-Vasic, “Very similar spacing-effect patterns in very different learning/practice domains,” *PLOS ONE*, vol. 9, no. 3, p. e90656, Mar. 2014, doi: 10.1371/journal.pone.0090656.